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Accurate and Robust Locomotion Mode Recognition Using High-Density EMG Recordings from a Single Muscle Group

Shibo Jing, *Student Member, IEEE*, Hsien-Yung Huang, *Member, IEEE*, Ravi Vaidyanathan, *Member, IEEE,* and Dario Farina, *Fellow, IEEE*

***Abstract* —Existing methods for human locomotion mode recognition often rely on using multiple bipolar electrode sensors on multiple muscle groups to accurately identify underlying motor activities. To avoid this complex setup and facilitate the translation of this technology, we introduce a single grid of high-density surface electromyography (HDsEMG) electrodes mounted on a single location (above the rectus femoris) to classify six locomotion modes in human walking. By employing a neural network, the trained model achieved average recognition accuracy of 97.7% with 160ms latency, significantly better than the model trained with one bipolar electrode pair placed on the same muscle (71.4% accuracy). To further exploit the spatial and temporal information of HDsEMG, we applied data augmentation to generate artificial data from simulated displaced electrodes, aiming to counteract the influence of electrode shifts. By employing a convolutional neural network with the enhanced dataset, the updated model was not strongly affected by electrode misplacement (93.9% accuracy) while models trained by bipolar electrode data were significantly disrupted by electrode shifts (29.4% accuracy). Findings suggest HDsEMG could be a valuable resource for mapping gait with fewer sensor locations and greater robustness. Results offer future promise for real-time control of assistive technology such as exoskeletons.**

I. Introduction

An accurate real-time locomotion mode recognition is one of the key elements for active assistive robotic interfaces to enable appropriate and safe control. Bipolar EMG has often been used for recognition and normally needs recordings from multiple muscle groups. For example, to discriminate five locomotion modes during steady state period, Huang et al. [1] measured EMG signals from 11 muscles to achieve an accuracy of around 99% and Hu et al. [2] recorded from 14 muscles in combination with bilateral IMUs and goniometers to achieve an accuracy of 99.2%. However, it often ends up with a large number of sensors mounted on the human body, which reduces user comfort and increases setup time.

HDsEMG features a large number of electrode channels within a limited surface area and within a single electrode grid, thereby offering spatial and temporal information on muscle activity. Because HDsEMG covers a larger skin surface than individual bipolar recording, it could detect more global activity on a muscle group than individual bipolar recordings. This may enable it to record distinct patterns for different locomotion modes that cannot be identified using bipolar EMG. Mounting a single HDsEMG system over a muscle group rather than locating several bipolar recording electrodes on multiple muscle groups would have the advantage of simpler donning of the interface as well as simpler integration into assistive devices, such as exoskeletons. HDsEMG has been successfully used in arm movement recognition to identify 8 gestures with an accuracy of 98.6% [3]. However, it has not yet been applied for lower limb motion recognition. In this study, we investigated the use of HDsEMG recorded from a single muscle for locomotion recognition.

We applied a HDsEMG grid on the rectus femoris muscle, which is one of the major active muscles during locomotion, to recognize six common locomotion modes: standing, level walking, stair ascent and descent, ramp ascent and descent. First, we analysed the accuracy between models trained using the entire HDsEMG recordings and models trained using a single or multiple pairs of bipolar EMG extracted from the same HDsEMG recordings. Afterwards, we investigated the influence of electrode shift on classification precision, as changes in electrode position substantially influence prediction reliability [4]. As HDsEMG offers substantial information to allow generating synthetic data, which is not possible for bipolar electrodes, we proposed a data augmentation approach to enhance classification robustness in the presence of electrode shifts. By creating artificial data incorporating potential electrode shifts, we hypothesised that the improved model is less affected by estimating error due to electrode shift.

II. Materials and Methods

A. Experiment setup

Seven healthy subjects (four males and three females, aged between 20 and 30 years, with no previous leg injuries) participated in the experiment. A 13 × 5 monopolar channel

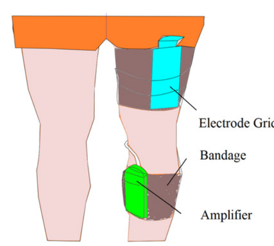

Fig. 1 Experimental set-up. The electrode grid was secured using surgical adhesive tapes and self-adhesive bandages to reduce movement during walking. The EMG amplifier was secured onto the fibula using self-adhesive bandages.

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S.Jing and R.Vaidyanathan are with the Department of Mechanical Engineering, Imperial College London, London, United Kingdom (e-mail: s.jing19@imperial.ac.uk, r.vaidyanathan@imperial.ac.uk).

HY.Huang and D.Farina are with the Department of Bioengineering, Imperial College London, London, United Kingdom (e-mail: h.huang14@imperial.ac.uk, d.farina@imperial.ac.uk).

Co-first author: S.Jing and HY Huang

Corresponding author: S.Jing

grid with an inter-electrode distance of 8mm (GR08MM1305, OT Bioelettronica, Italy) was mounted on the rectus femoris muscle of the left leg to record HDsEMG signals (Fig.1). The signals were sampled at 2000Hz, amplified by a 64-channel amplifier (Sessantaquattro, OT Bioelettronica, Italy) and recorded by a laptop wirelessly through WIFI communication. Before electrode placement, the skin area was shaved and cleansed using abrasive paste and alcohol. Conductive cream was applied onto the grid to ensure satisfactory contact between the electrodes and the skin. The electrode placement followed the guideline of SENIAM [5]. The experimental protocol was approved by the Imperial College Research Ethics Committee (ethical approval number: 18IC4685).

B. Experiment Protocol

The entire experiment comprised standing, level-ground walking, stair ascending and descending, ramp ascending and descending. Each subject was first asked to quietly stand upright for 30 s and then to walk along a 15-m flat runway. When they reached the end, they were asked to stand for 3 s, turn around and stand for another 3s before walking back. This was considered as a cycle and was repeated twice. For the stair ascending and descending task, a staircase consisting of 15 steps with a step height of 17 cm was used. The subjects waited for 3 s before starting stair ascending and 3 s before stair descending, repeating this cycle six times. The ramp ascending and descending task was performed on a 20-m ramp with an angle of approximately 6 degrees. These tasks were performed with the same waiting intervals as the other tasks and each cycle of ascending/descending was repeated twice. During the experiments, a metronome (set to 100 BPM) was held by an experimenter walking along with the subject to maintain a consistent cadence. Breaks were allowed between cycles to avoid fatigue. Each task was completed within 10 min and therefore muscle fatigue was minimal.

C. Signal Processing

1) Signal Filtering

A 4th order Butterworth bandpass filter was used with cut-off frequencies of 20Hz~400Hz. One channel at the corner of the electrode grid is missing due to the design. Any missing channels (with zero value recorded throughout the experiment) were replaced by using the average value of the electrodes surrounding them. No more than two channels were found missing in all experiments. Some channels within the array could occasionally disconnect during the experiments. To mitigate this effect, a two-dimensional median filter with kernel size of 3 was adopted for compensation.

2) Feature Extraction

After filtering, the data were cropped to extract the steady-state walking sequence, for which the data from the turning or transition period during the experiment were discarded. Afterwards, sliding analysis windows were used to segment the EMG data. The length of the sliding window was 256 ms, and the window increment was 32 ms. Eight time-domain features were extracted from each window: mean absolute value (MAV), root mean square (RMS), waveform length (ML), slope sign changes (SSL), zero-crossings (ZC) and third order autoregressive coefficients. These features were normalized to have zero mean and unit variance, and then used as input to the classification model.

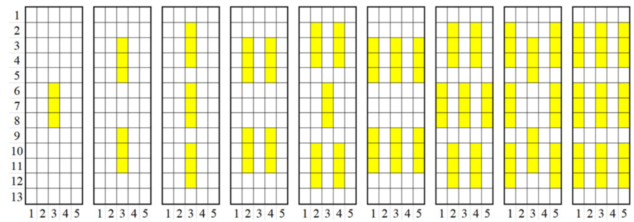

Fig. 2 Selected bipolar EMG combinations. Each square refers to a monopolar electrode pair on the HDsEMG grid. Each yellow block covering three squares refers to a bipolar selected, where the first and third electrodes were used to calculate the bipolar signal. Due to the grid size, the largest number of bipolar EMG that can be selected is nine.

3) Bipolar EMG selection

In order to compare the performance between HDsEMG and bipolar EMG derivations on the same muscle, we selected several bipolar combinations from the HDsEMG grid. Each bipolar EMG was derived from the same electrode grid by taking the monopolar voltage difference between two electrodes. The bipolar EMG selection scheme was based on the concept of keeping the position of bipolar EMG channels evenly distributed on the muscle/gird. Considering that common bipolar electrodes have a diameter of 10-20mm, the bipolar derivations extracted from the grid had an inter-electrode distance (IED) of 16 mm, as shown in Fig. 2.

4) Electrode shift

Transversal electrode shift of 1 IED, referring to 8mm displacement, was investigated in this study (Fig.3), as transversal shift has a larger impact on signal quality than longitudinal shift [6]. As shown in Fig.3.A, the HDsEMG signals collected from the area marked in yellow were selected as pre-shift data, while the HDsEMG signals collected from the green area were selected as after-shift data.

For comparison, an 8-mm electrode shift was also applied to the bipolar EMGs. We selected four cases of electrode shifts. Fig.3.B.(a) refers to the selected bipolar EMG combination before electrode shift, which covers the same skin area as the HDsEMG shown in Fig.3.A.(a). The next four images (Fig.3.B.(b)-(e)) respectively show the example of one, three, four, and six electrode shifts, marked with green colour.

5) Data augmentation and interpolation

The high number of electrodes in HDsEMG presents richer spatial information than bipolar EMG pairs. EMG values after spatial displacement can possibly be inferred in advance and included in the dataset for model training, which is not applicable to regular bipolar EMG pairs. This technique is

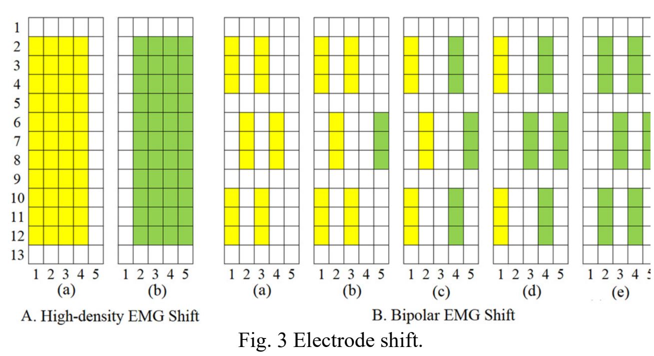

Fig. 3 Electrode shift.

called data augmentation and may help improve the recognition performance against spatial displacement.

In order to augment data with displacements that are a fraction of the IED, HDsEMG signals were spatially interpolated. The requirement for reliable interpolation is that the IED should be lower than 10 mm [7], which is satisfied in our study. The bicubic interpolation method was used such that image resolution was increased from 13 × 5 × 8 to 97 × 33 × 8, with the pixels corresponding to the 97mm × 33mm area covered by the electrode grid.

D. Locomotion Mode Classification Model

Two classification models were used, with training performed on MATLAB (2021a) deep learning toolbox.

1) Three-layer neural network

A three-layer neural network with the same structure was applied to HDsEMG and bipolar EMG for locomotion classification. This network consists of three hidden layers with 200 hidden units for each. Each hidden layer is followed by a batch normalization layer, a ReLU layer and a dropout layer (dropout rate of 50%). It is then connected to a full connected layer of size 6 and a softmax layer for output.

For either HDsEMG dataset or bipolar EMG dataset, the first 80% of the dataset was used as the training set, the middle 10% was used as validation set for hyper-parameters tuning and the last 10% was used as the testing set. The training set was shuffled while the validation set and the test set were not shuffled. All training and testing process were repeated five times to obtain the average value. When calculating the performance under electrode shift, the test set was shifted as described in section C.4, with the training set unchanged.

2) Convolutional neural network (CNN)

To apply the data augmentation method on HDsEMG, EMG image interpolation was needed, and it increased the dataset size significantly. Considering that CNN has high performance in image processing and runs much faster than a normal neural network, a CNN model was employed here on the HDsEMG dataset as described in section C.4, to calculate the accuracy after using data augmentation.

The CNN model consisted of a convolutional layer with a kernel size of 5 × 5 × 30 and a dilation factor of 2 × 2, followed by a batch normalization layer, a ReLU layer, an average pooling layer (pool size of [2 2] and stride of [2 2]) and a dropout layer (dropout rate of 50%). Then it is connected to a full connected layer of size 6 and a softmax layer for output.

In the CNN model, all HDsEMG feature images in the dataset were interpolated first. Then the first 80% of the dataset was individually shifted transversally for 1~8mm randomly with a repetition of three times. This process created artificial EMG data covering most of the potential electrode displacement. The newly created data were then combined with the original training set (i.e., the first 80% of the original dataset) to construct an augmented training set (four times the size of the original training set). To compare the result before and after data augmentation under electrode shift, the last 10% of the original dataset was shifted as described in section C.4 to form the testing set with shifted data.

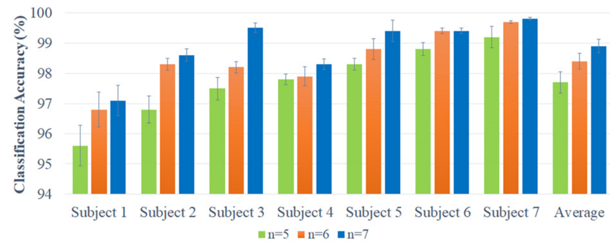

Fig. 4 Classification accuracy of each subject using HDsEMG for six locomotion modes. n is the majority voting number. Error bars represent the standard error.

Finally, for both models, a majority vote method was used to post-process the classification decisions [8]. For a given point, the previous n samples and the next n samples were considered to make a final decision. The class with the greatest number of occurrences in the 2n+1 points was identified as the current locomotion mode. In this study, values for n of (five, six and seven) were tested.

E. Evaluation

Recognizing multiple locomotion modes can be regarded as a multiclass classification problem. Accuracy was used as the evaluation metric:

$$Accuracy = \frac{number\ of\ correctly\ classified\ test\ label}{total\ number\ of\ test\ label} \times 100\% \quad (1)$$

III. Results And Discussion

A. Accuracy using high-density EMG

As seen in Fig. 4, when the majority voting number n was selected to be 5, the average recognition accuracy was 97.7%, ranging from 95.5% to 99.2% across subjects. As the window increment in our study was 32ms, the corresponding delay was 160ms. The sample confusion matrix derived from n =5 is shown in Table 1. The ramp ascent mode is the most difficult scenario to be recognised and can easily be misclassified as ramp descent. With the increase of the majority voting number n, a better result could be obtained, with a longer delay. When n = 6, the average accuracy was improved to 98.3% and for n = 7, the average accuracy achieved 98.8% (Fig. 4).

B. Accuracy using selected bipolar EMG

The majority voting number n was set to 5 when we classify the locomotion mode using selected bipolar EMG combinations.

TABLE I. Confusion Matrices of N = 5

Target \ Predict	Standing	Level walking	Stair Ascent	Stair Descent	Ramp Ascent	Ramp Descent
Standing	**100.00**	0.00	0.00	0.00	0.00	0.00
Level walking	0.00	**98.23**	0.26	0.00	0.72	0.79
Stair Ascent	0.00	0.39	**97.16**	1.82	0.27	0.36
Stair Descent	0.00	0.54	1.26	**97.83**	0.00	0.38
Ramp Ascent	0.00	0.19	0.30	0.23	**96.20**	3.08
Ramp Descent	0.00	0.36	0.12	0.54	0.97	**98.02**

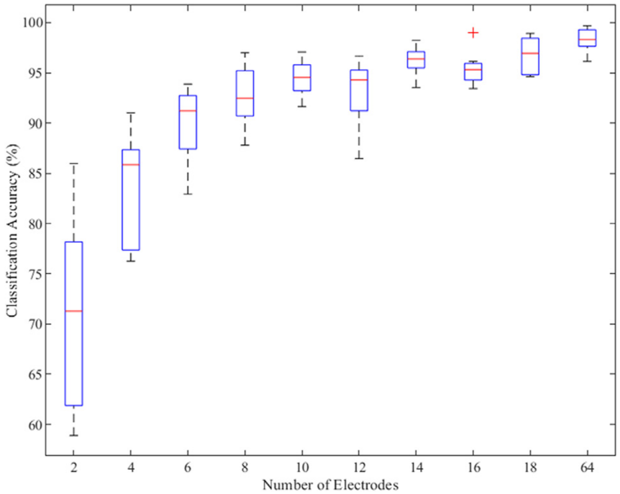

Fig. 5 Classification accuracy using different number of electrodes. Two electrodes form one bipolar pair with the selection method described in Fig. 2. The '+' markers the outlier.

The accuracy increased in general as the number of bipolar EMG increased (Fig. 5). The lowest prediction accuracy was obtained in the case of using only one pair of bipolar electrodes (median accuracy 71.4%). To reach a similar level of 97.8% median accuracy as HDsEMG (the 64 electrodes in Fig. 5), nine bipolar electrodes (18 electrodes) were needed (96.5%).

The variance of classification accuracy across subjects generally decreased as the number of electrodes increased, particularly when the number of electrodes increased from 2 to 10. When only one bipolar EMG was used, the accuracy varied significantly from <60% to >85%. However, even the best result is much worse than the result using HDsEMG on the same muscle. When the number of electrodes increased to 10, the accuracy ranged in 91.5%-97% across subjects.

Another observation was that more bipolar electrode pairs did not always guarantee better accuracy. Fig. 5 shows that the model trained with 16 electrodes performed worse than with 14 electrodes, possibly due to electrode position selection. It also implies that we need to carefully select the best position of bipolar EMG combinations to achieve better performance. However, the selection process requires systematic investigation, which can be time consuming and sensitive to various physiological factors. In contrast, the result of using HDsEMG is shown to be more stable and less sensitive to electrode position compared to using bipolar EMG, so it might be a better solution as less work is required for robust results.

C. *Accuracy after electrode shift*

Electrode shift had a significant impact on classification results (Fig. 6). A shift of 1 IED in a single bipolar derivation (as shown in Fig.3.B(b)) resulted in an accuracy drop from 95.2% to <90%. If half of the bipolar EMG shifts (Fig.3.B(c)), the accuracy would be lower than 50%. When all bipolar EMG shift (Fig.3.B(e)), the accuracy is only 29.4%.

For HDsEMG, full electrode shift (as shown in Fig.3.A(b)) caused the accuracy to decrease from 96.2% to 32.4%. However, it could be compensated back to 93.9% by using the data augmentation method. This suggested that HDsEMG offers the possibility of using data augmentation to create a robust prediction model that can counteract electrode displacement. When using the data augmentation method,

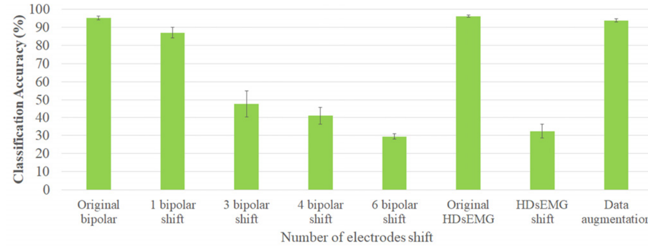

Fig.6 Classification accuracy after electrodes shifting for both bipolar EMG and HDsEMG. The accuracy presents the mean value across seven subjects while the error bars represent the standard errors.

there is a trade-off between the number of artificial data, computation burden and resulting improvement. In our study, it was found that three times were the best number of repetitions as more repetitions offer little benefit.

IV. Conclusion

The high spatial resolution offered by HDsEMG does provide additional valuable information to allow high recognition accuracy when only one muscle is recorded (97.7% accuracy with 160 ms latency), in comparison with one bipolar electrode pair placed on the same muscle (71.4%). Furthermore, through data augmentation, HDsEMG shows the potential of exploiting redundant signals to improve model robustness against spatial displacement with a small reduction in accuracy of around 2% for up to 8 mm displacements. By adding a small number of additional muscles, it is expected that using HDsEMG approach would further improve the recognition accuracy. In addition, extra experiments are to be conducted in the future to compare the performance of using HDsEMG on a single muscle with using bipolar EMG on multiple muscles.

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